

The Origins of the STEM Gender Gap: Evidence from an Analysis of Secondary School Subject Selection

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Abstract

Secondary school subject choices are a key gateway to STEM (Science, Technology, Engineering and Maths), yet their determinants remain underexplored. Using the Growing Up in Ireland study, this paper employs regression and decomposition analysis to examine STEM participation. We document substantial gender gaps in practical-based STEM subjects, but not in science or maths. Lower secondary choices play a central role in shaping later trajectories and explain most upper secondary gender gaps. However, lower-secondary gaps are not explained by conventional mechanisms such as early characteristics, comparative advantage, schools, or peers. Results are robust to alternative specifications, including machine learning methods.

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STEM fields play a critical role in driving economic growth, innovation, and productivity, contributing substantially to national competitiveness. It is well established that substantial gender disparities exist and persist in STEM participation in higher education. These disparities have important implications for gender equity, wage equality, and diversity in innovation-driven sectors, making them an important focus for policymakers.¹ As a result, a large literature has examined the determinants of gender gaps in STEM participation in higher education. A consistent finding across this literature is that secondary school subject choices strongly predict later STEM pathways (Card and Payne, 2021; Chevalier, 2011; Conger, Kennedy, Long and McGhee, 2021; Delaney and Devereux, 2019; Gottfried and Bozick, 2016; Granato, 2023; McNally, 2020; Tchuente, 2016). This points to secondary school as a critical stage at which gender differences in STEM pathways emerge, and therefore a key setting for understanding the origins of these gaps.

Treating the “STEM gender gap” as a single phenomenon risks obscuring important differences across subjects. We find that in secondary school, female students are more likely than male students to take traditional academic STEM subjects such as biology and chemistry, while large gender gaps in favour of male students arise primarily in practical, technology-oriented subjects. This suggests that the relevant question is not simply why girls avoid STEM, but rather which types of STEM fields exhibit gender gaps, and at what stage those gaps emerge.

Existing studies of secondary school subject choice have highlighted several possible explanations for gender differences, including psychological factors and preferences, such as self-confidence and appreciation for STEM subjects (Stokking, 2000), and broader psychological and social influences (Mackenzie, Berger and Holmes, 2021). School-related attributes and teaching practices also influence enrolment decisions (Smith and Hannan, 2006). Schools may matter through peer effects: female students are more likely to select STEM subjects when exposed to high-achieving peers of the same sex (Mouganie and Wang, 2020), whereas exposure to a larger share of girls in earlier grades can reduce the likelihood of choosing traditionally female-dominated subjects (Schneeweis and Zweimüller, 2012). Finally, some research has emphasised the role of sex-specific comparative advantage, suggesting that gender gaps might partially reflect expected probabilities of success across different subjects (Delaney and Devereux, 2019; Goulas, Griselda and Megalokonomou, 2024). However, prior maths achievement alone does not fully explain these patterns (Friedman-Sokuler and Justman, 2016). Most existing literature focuses on the later stages of secondary school, examines traditional STEM subjects such as science and mathematics, and emphasises a single theoretical perspective.

This study contributes to this literature in two ways. First, we show that gender gaps in secondary school STEM participation are highly uneven across subjects. In particular, large gaps are concentrated in practical,

technology-oriented subjects, such as technology, engineering, construction, and design and communication graphics, whereas gaps in traditional academic STEM subjects are small or reversed. These practical subjects are often overlooked despite their economic importance and direct pathways to high-demand, high-wage STEM-related careers. Taking these subjects has also been found to significantly positively predict the likelihood of applying for a STEM college degree (Delaney and Devereux, 2019). Second, we examine how these gender gaps evolve across secondary school, tracing subject choices from lower secondary through to upper secondary education. This allows us to analyse how early subject choices persist and shape later educational pathways.

We first document substantial gender gaps in enrolment in practical STEM subjects at the upper secondary level. These gaps are strongly related to earlier subject choices: participation in STEM subjects at lower secondary school, which we interpret as a measure of “STEM readiness”, explains a large share of the later disparity. We therefore turn our attention to the lower secondary level and show that gender gaps are already pronounced at this early stage. These differences persist even after controlling for prior academic performance and a rich set of observable characteristics measured at age 9, indicating that the divergence in STEM pathways cannot be explained by the early-life factors we observe. Conventional explanations for gender gaps in STEM, including comparative advantage, school environments, and peer composition, account for only a small portion of the observed disparity in practical STEM subjects. We also show that the results are not driven by differences in subject availability across schools, an often-cited explanation for gender gaps in STEM, nor by gender sorting across schools; in fact, the gap is larger within coeducational schools than in single-sex schools. Finally, to ensure that the findings are not driven by functional form assumptions or variable selection, we conduct a robustness exercise using flexible machine learning methods and a substantially expanded set of explanatory variables. This approach allows for rich nonlinearities and high-dimensional interactions, making a much broader set of potential predictors available to the model. The results remain largely unchanged, increasing confidence that the early divergence in practical STEM participation is not explained by the observable characteristics available in our data.

The rest of the paper is organised as follows. [Section 2](#) outlines the Irish education system, the Growing Up in Ireland data, and our methodological approach. [Section 3](#) presents our results for upper secondary STEM enrolment, including regression analysis and decomposition of the gender gap. [Section 4](#) turns to lower secondary subject choices and applies the same analytical framework. [Section 5](#) addresses robustness, focusing on the potential role of school choice in driving results, particularly given the prevalence of single-sex schools in Ireland, and on the use of flexible machine learning methods to ensure that the persistence of

large unexplained gaps is not driven by restrictive modelling assumptions. [Section 6](#) concludes.

I. INSTITUTIONAL BACKGROUND, DATA AND METHODOLOGY

This research draws on data from the Growing Up in Ireland (GUI) Cohort '98, a nationally representative longitudinal study tracking the lives of Irish children. The study's first wave surveyed 8,568 children aged 9 from 910 randomly selected primary schools between August 2007 and May 2008. The sample accounts for roughly 14 per cent of Ireland's nine-year-old population at the time. Data collection involved structured interviews with the children, their parents or primary caregivers, teachers, and principals. Unlike administrative data, the GUI dataset contains rich information on students' backgrounds and subject choices across schooling levels, making it well-suited to examining the early determinants of STEM subject selection.

Secondary education in Ireland begins around age 12 and is divided into two distinct phases, each ending with a national examination. The first phase, known as the Junior Cycle (i.e. lower secondary), spans three years and concludes with the Junior Certificate (JC). Students then progress to the Senior Cycle (i.e. upper secondary) around age 15, with this cycle lasting two to three years and culminating in the Leaving Certificate (LC), the primary qualification for higher education entry. To give international context, the Junior Cycle is roughly equivalent to middle school in the U.S. education system, while the Senior Cycle corresponds to high school. On average, students take ten to eleven subjects for the JC and seven to eight for the LC. Throughout both phases, all students are required to study Irish, English, and Maths, with occasional exemptions from Irish granted for students with learning disabilities or those who relocated to Ireland after the age of 12. The remaining subject choices are determined by school availability and student interest. Maths and Irish are offered at higher, ordinary, or foundation levels, while English and all non-compulsory subjects are available at higher or ordinary levels. JC grades are letter-based, whereas LC grades are converted to a common points scale used to determine admission to post-secondary courses. A maximum of 56 points can be earned by taking a subject at ordinary level (by achieving the top grade of Ordinary-Level 1) and 100 in taking a subject at higher-level (by achieving the top grade of Higher-Level 1). Ordinary level subjects had a maximum of 60 points prior to a 2017 grading system reform, designed to reward higher-level participation while maintaining overall point fairness. Since 2012, a 25-point bonus has been awarded for achieving a passing grade in higher-level maths.

The Junior Cycle offers a broad curriculum with limited subject choices. The Senior Cycle offers greater specialisation through a wider range of subjects. For instance, the general science subject studied in the Junior Cycle becomes specialised in physics, chemistry, and biology in the Senior Cycle. Students retain the autonomy to choose their examination level for each subject. Access to the majority of higher education courses is determined solely by LC performance. Third-level institutions commonly set subject-specific entry criteria; for example, Economics (BSc) at University College Dublin requires a Higher-Level grade 5 in Maths and an Ordinary-Level grade 6 in English. Among applicants who meet the subject-level and grade requirements, the highest-scoring individuals on the common points scale gain entry. Consequently, the subjects students choose in secondary school play a critical role in shaping access to higher education, as they determine whether students meet course prerequisites and the number of points they can accumulate for admission to college programmes.

Ireland provides a particularly informative setting for studying gender differences in STEM subject choice, due to the continued prevalence of single-sex secondary schools. Single-sex schools remain a prominent feature of the Irish education landscape, with around 50% of students enrolled in them. In contrast to many other countries, these schools do not typically serve demographically distinct populations; their continued prevalence reflects historical legacy rather than deliberate sorting. Most have retained their original structures since their founding, when single-sex education was the norm.

I.1 INDEPENDENT VARIABLES

Recognising that secondary subject choices likely reflect multiple underlying factors, we exploit the unusually rich covariate information in the Growing Up in Ireland (GUI) survey to construct detailed measures of the main mechanisms proposed in the literature on gender gaps in STEM participation. Our approach is informed by the extensive literature that primarily examines gender and higher education STEM enrolment, from which we borrow several theoretical frameworks (see [Delaney and Devereux \(2021\)](#) for a review of this literature and the potential theoretical mechanisms).

Specifically, we examine four distinct theoretical mechanisms that have been highlighted as explaining gender gaps in STEM enrolment. The first mechanism concerns “STEM-readiness”. The primary motivation for examining upper secondary STEM subject choices is the recognition that these choices directly influence STEM-readiness in higher education. However, it is possible that existing gender gaps at the upper secondary level already reflect earlier differences in STEM-readiness, potentially stemming from subjects studied at the lower secondary level. To address this, we include indicators for each optional JC subject chosen, indicators for whether students took mandatory subjects (maths and English) at a higher level, and an indicator for

whether the student completed Transition Year (an optional one-year programme with a focus on personal and social development taken after the JC, before proceeding to the Senior Cycle).

The second theoretical mechanism is comparative advantage. Evidence consistently shows that female students outperform male students in reading, while males have a relatively small advantage in maths. Although the sources of these differences (biological, cultural, or stereotype-driven) are debated, their existence frequently explains part of the STEM gender gap (Delaney and Devereux, 2019; Speer, 2017). Thus, we incorporate multiple cognitive and non-cognitive skill measures from the GUI dataset. Cognitive skills are measured at age 9 using standardised Drumcondra Verbal and Numerical Reasoning test scores. These standardised tests are administered to students nationally at multiple points throughout their primary education. These assessments are designed in line with the Irish educational curriculum and evaluate students' proficiency in English (verbal reasoning) and Maths (numerical ability). Further, JC examination results, particularly grades in maths and English and the number of JC subjects taken at higher-level are included as direct proxies of academic ability.² One key strength of our data is that we can also include measures of non-cognitive skills, which have been shown to explain gender disparities in educational choices at the secondary level (Buser, Peter and Wolter, 2017). In particular, we include scores from the Strengths and Difficulties Questionnaire (SDQ), a measure of conscientiousness from the Ten Item Personality Inventory, and a measure of academic self-concept from the Piers-Harris scale.³ The latter is essentially a measure of a student's own confidence in their academic ability, which has previously been proposed as a potential mechanism underlying gender gaps in achievement in STEM (Di Tommaso, Mccagnan and Mendolia, 2021).

Third, we consider school factors and peer effects. A common focus of the literature is the effect of single-sex schooling on the propensity of female students to take STEM courses (Calkins, Binder, Shaat and Timpe, 2023). Our study's Irish context is particularly interesting given the large presence of single-sex secondary schools. While our data do not permit direct identification of peer effects, we include extensive information about school characteristics that could influence STEM enrolment decisions. School characteristics include subject offerings; a key advantage of our data is that we observe the full set of subjects offered by each school, allowing us to directly control for subject availability, a factor frequently highlighted in the literature on gender gaps in STEM. Other school-level controls include school type (e.g., fee-paying or non-fee-paying) and whether the school is designated as a "DEIS" school (i.e., one that serves a high proportion of students at risk of educational disadvantage). To further approximate peer influences, we explicitly include indicators of single-sex versus mixed-gender school environments at the secondary level.

Fourth, gender gaps in STEM enrolment may reflect fundamental differences in preferences regarding

fields of study or related career paths. For instance, [Kuhn and Wolter \(2022\)](#) highlights that female students tend to favour fields involving interpersonal interactions, a description less applicable to many practical STEM subjects. Such differences in preferences may stem from societal norms or gender stereotypes. While we do not directly observe preferences for specific job characteristics, the dataset contains rich information on students’ interests, attitudes, and aspirations. In particular, we observe students’ stated occupational aspirations, which we categorise by broad employment field. We also incorporate measures of students’ enjoyment of maths and reading, whether their stated favourite subject in the JC is practical, and whether they report liking school. Finally, we account for students’ perceptions of the difficulty of maths and English, and their expectations regarding their own educational attainment (how far they expect to go in their educational journey).

Finally, we incorporate a range of background variables. While these are not expected to directly explain gender gaps, they may nevertheless influence overall STEM enrolment and, if omitted, could confound the analysis. These variables include the primary caregiver’s immigration status (born in Ireland or not) and education (university-level educated or not), the number of books in the home, and health-related measures, such as whether the young person has a disability.

Starting with an initial balanced panel of 6,039 students observed across Waves 1 to 3, the sample size is reduced primarily due to missing data on key variables central to the analysis. In particular, we exclude students who did not report their JC maths or English level or grade, as well as students attending schools for which the principal did not complete the school characteristics survey in Waves 2 or 3. These exclusions account for the majority of the reduction in sample size and are necessary given the paper’s core focus on subject-level outcomes and school characteristics. After applying all sample restrictions, the final analytical sample consists of 3,888 students. See Appendix Table A.2 for a breakdown of observations lost through our sampling restrictions. In total, we use 58 independent variables, which we group based on the mechanisms listed above: STEM-readiness/Junior Cert, comparative advantage/skills, peer effects/school factors, preferences/tastes, and background characteristics. These core variables are selected *ex ante* because they map transparently onto the main mechanisms emphasised in the literature, and can therefore be grouped in an interpretable Gelbach decomposition. This comes at the cost of excluding some additional variables that may contain predictive information but do not fit cleanly into these mechanism groups, are measured less directly, or would substantially reduce the transparency of the main specification. We use these additional variables in the machine learning analysis in [Section 5.1](#). [Table 1](#) provides summary statistics for a subsample of variables from each grouping. A full list of variables in this analysis is in Appendix [Section 7.3](#). A similar

set of variables and dataset is used by [Crotty and Murphy \(2025\)](#).

I.2 DEPENDENT VARIABLE

The optional LC STEM subjects can be grouped into four categories: science subjects (Biology, Chemistry, Physics, Agricultural Science), technology subjects (Technology, Design and Communication Graphics (DCG)), engineering subjects (Engineering, Construction Studies), and mathematics subjects (Maths, Applied Maths). Two other STEM subjects currently on the LC curriculum are excluded from our analysis: Computer science, which was not offered as a subject when our sample sat their LC exams, and a combined "Physics & Chemistry" subject, which is offered by few schools and therefore taken by an extremely low number of students in our sample and nationally. The STEM subjects listed above vary markedly from each other. Technology, DCG, construction, engineering and agricultural science feature substantial practical components (for example, designing and constructing a product for engineering or submitting a report on practical farm experience for agricultural science), while the other subjects (maths, applied maths, biology, chemistry, and physics) are more traditional academic subjects and are solely assessed by a written final exam. As a result, one might expect different factors to influence enrolment in these two subject groups.

Table 1: Summary Statistics - Independent Variables

Variable	Mean	Max	Min
STEM Readiness/Junior Cert			
Studied Technical Graphics for JC	0.20	1.00	0.00
Studied Metalwork for JC	0.10	1.00	0.00
Studied Music for JC	0.20	1.00	0.00
Studied Foreign Language for JC	0.90	1.00	0.00
Comparative Advantage/Skills			
Drumcondra Reading Score	0.12	2.87	-3.36
Drumcondra Numerical Score	-0.65	1.90	-3.62
Number of Higher-Level Subjects Taken for JC	8.21	12.00	0.00
Academic Self-Concept Score	12.10	16.00	0.00
Peer Effects/School Factors			
Attended DEIS School	0.15	1.00	0.00
Attended Fee-paying School	0.11	1.00	0.00
Agricultural Science Offered by School	0.66	1.00	0.00
Attended All Boys Secondary School	0.23	1.00	0.00
Preferences/Tastes			
Favourite Subject for JC is Practical Subject	0.18	1.00	0.00
Young Person (YP) Likes Maths	1.61	3.00	1.00
YP Likes School	1.81	3.00	1.00
YP Preferred adult occupation – Healthcare	0.11	1.00	0.00
Background			
Primary Caregiver (PCG) is University Educated	0.18	1.00	0.00
YP has a Disability	0.10	1.00	0.00
Number of Books in home for YP	4.25	5.00	1.00
PCG born outside of Ireland	0.15	1.00	0.00

Notes: N=3,888. YP = young person (survey child); PCG = primary caregiver (typically mother). All statistics are computed using survey weights. Binary variables are coded as indicators taking values 0/1. Ordinal variables (“Likes Maths”, “Likes School”) are measured on a 1–3 scale, 1=Always, 2=Sometimes, 3=Never. “Number of Books in Home” is measured on a 1–5 categorical scale. A full list of variables is provided in Appendix Table A.4.

Table 2 presents the percentage of students of each gender enrolled in each subject of interest in our sample. Enrolment rates at the national level are provided in Appendix Section 7.2 to show our data is nationally representative. Clear gender differences emerge in subject selection patterns. Despite widespread discourse highlighting the underrepresentation of females in STEM, Biology and Chemistry have higher enrolment rates among female students than male students. While gender gaps favouring male students are also evident in Physics and Applied Maths, these are relatively modest compared to the substantial disparities observed in practical STEM subjects. Using the same dataset, Crotty and Murphy (2025) show that substantial disparities exist in enrolment rates in academic STEM subjects along the lines of socioeconomic status, not gender. Large gender gaps favouring male students are observed in construction studies (22.22%

of boys versus 1.82% of girls), design and communication graphics (DCG) (15.57% of boys versus 3.57% of girls), and engineering (14.21% of boys versus 0.74% of girls). The gender gap in agricultural science is smaller but still substantial (14.26% of boys versus 9.42% of girls). Technology has the lowest overall enrolment rate in our sample, with 3.38% of total students taking the subject, but there is nevertheless a substantial gender gap (5.58% of boys versus 1.06% of girls).

Despite their prevalence in the curriculum, practical STEM subjects are often overlooked in studies of STEM gender gaps. However, recent evidence underscores their importance. Using administrative data, [Delaney and Devereux \(2019\)](#) shows that, among college applicants, taking any of these five practical subjects significantly increases the probability of enrolling in a STEM course in higher education in Ireland. [Delaney and Devereux \(2019\)](#) finds that all five of these subjects have a statistically significant positive effect on the likelihood of applying for a STEM degree. Strikingly, four of the five practical subjects (Technology, DCG, Engineering, and Construction Studies) are stronger predictors of STEM enrolment than Biology, the most popular STEM subject at LC level. Only Chemistry, Physics, and the rarely taken combined Physics & Chemistry subject outperform Engineering, Technology and DCG as predictors of STEM enrolment in college. This evidence highlights the economic relevance of understanding who takes practical STEM subjects. Given their importance as predictors of post-secondary STEM enrolment, the substantial gender gaps in participation, and the limited existing research, we focus our analysis on practical STEM subjects. A brief description of these subjects is given below.⁴

Both technology subjects place strong emphasis on the use of information technology. Design and Communication Graphics (DCG) emphasises spatial reasoning, graphical communication, and CAD (computer-aided design). Core areas of study include plane and descriptive geometry and information and communication technologies. Technology focuses on solving technological problems, with core topics including information and communication technologies, energy, electricity and electronics, and communications and graphic media. The construction syllabus explicitly aims to provide a basis for those who want to study construction technologies at the third level, and it therefore combines theoretical knowledge of construction principles with practical workshop and laboratory skills. Similarly, engineering studies a wide range of mechanical engineering materials, processes and technological applications. Agricultural science is the study of the scientific and technological principles and practices underlying agriculture. Topics include plant and animal biology, soil science, and sustainability, emphasising experiential learning via the submission of a “Practical Experience Report.” While the specific subject titles are unique to the Irish curriculum, these subjects closely correspond to applied STEM subjects offered in many other education systems.

To incorporate all of the practical subjects listed above, we construct a composite outcome variable called *Practical STEM*, which equals 1 if a student studies at least one of these subjects and equals 0 otherwise.⁵ 45.96% of the 1,906 male students in our sample take at least one practical STEM subject at Leaving Cert level (and therefore have a value *Practical STEM* = 1), compared to 15.63% of the 1,982 female students. This gives us a gender gap of 30.33 percentage points. Throughout the paper, gender gaps are calculated as the difference in weighted conditional means between male and female students, with a positive gender gap indicating that a lower proportion of female students enrol than male students. Sampling weights are used throughout our analysis to address concerns regarding selective attrition due to our use of survey data. All figures presented throughout are weighted, and running the analysis without weights yields similar results. These gaps far outweigh those observed in the traditional academic STEM subjects on which the existing literature tends to focus. If we were to define a similar binary variable *Academic STEM*, which equals 1 if a student takes at least 1 of the subjects biology, chemistry, physics or applied maths and 0 otherwise, the gender gap is reversed in favour of girls at -6.41%.

Table 2: LC STEM Enrolment by Gender

Subject	Male (%)	Female (%)	Total (%)
<i>Non-Practical STEM Subjects</i>			
Biology	48.88	69.03	58.70
Chemistry	15.89	20.09	17.94
Physics	24.94	8.66	17.01
Applied Maths	8.07	3.11	5.65
<i>Non Practical STEM</i> (1+ non-practical subject)	70.46	76.86	73.58
<i>Practical STEM Subjects</i>			
Construction Studies	22.22	1.82	12.28
Agricultural Science	14.26	9.42	11.90
Design and Communication Graphics	15.57	3.57	9.72
Engineering	14.21	0.74	7.64
Technology	5.58	1.06	3.38
<i>Practical STEM</i> (1+ practical subject)	45.96	15.63	31.17
Total Observations	1,906	1,982	3,888

Notes: Percentages report weighted enrolment rates by gender for Leaving Certificate (LC) STEM subjects; for example, 48.88% of male students, 69.03% of female students, and 58.70% of all students in the sample study biology for LC. Composite rows (“Practical STEM” and “Non-Practical STEM”) indicate whether a student is enrolled in at least one subject within the respective category.

I.3 METHODOLOGY

We implement the decomposition within a linear probability model (LPM) framework. Let i index students and let Practical STEM_i denote an indicator for whether student i chooses a Practical STEM subject at upper secondary level. We begin with a baseline specification that captures the raw gender gap:

$$\text{Practical STEM}_i = \alpha + \beta \text{Female}_i + \varepsilon_i, \quad (1)$$

where Female_i is an indicator equal to one for female students. The coefficient β represents the unconditional gender gap in Practical STEM take-up. We then estimate a specification that includes a comprehensive set of covariates:

$$\text{Practical STEM}_i = \alpha + \beta^F \text{Female}_i + \mathbf{X}_i^{SR} \boldsymbol{\gamma}^{SR} + \mathbf{X}_i^{SP} \boldsymbol{\gamma}^{SP} + \mathbf{X}_i^{TP} \boldsymbol{\gamma}^{TP} + \mathbf{X}_i^{CA} \boldsymbol{\gamma}^{CA} + \mathbf{X}_i^{BG} \boldsymbol{\gamma}^{BG} + \varepsilon_i, \quad (2)$$

where $\mathbf{X}_i^{SR}$ captures STEM-readiness, $\mathbf{X}_i^{SP}$ captures school and peer effects, $\mathbf{X}_i^{TP}$ includes measures of tastes and preferences, $\mathbf{X}_i^{CA}$ captures comparative advantage, and $\mathbf{X}_i^{BG}$ contains background characteristics. The coefficient β^F therefore represents the conditional gender gap after accounting for these observable characteristics, where the superscript F denotes the full-specification estimate. All models are estimated using ordinary least squares, and standard errors are clustered at the school level to account for within-school correlation in outcomes.

To quantify the contribution of each group of covariates to the reduction in the gender coefficient between equations (1) and (2), we apply the decomposition proposed by [Gelbach \(2016\)](#). This approach exploits the omitted variable bias formula to express the change in the gender coefficient as the sum of contributions from each covariate group:

$$\beta - \beta^F = \Delta^{SR} + \Delta^{SP} + \Delta^{TP} + \Delta^{CA} + \Delta^{BG}. \quad (3)$$

Each Δ^g represents the portion of the change in the gender coefficient attributable to covariate group g . The Gelbach decomposition provides a nested and path-independent way of allocating the reduction in the gender gap across groups of explanatory variables.

II. RESULTS - UPPER SECONDARY

This section presents an analysis of enrolment in practical STEM subjects at the LC level. The analysis proceeds in two steps. First, we present regression results that describe which factors are associated with

practical STEM take-up at upper secondary school. Second, we focus explicitly on gender differences: we quantify the unconditional gender gap, examine how it changes when conditioning on the full set of covariates, and apply a Gelbach decomposition to attribute changes in the gender coefficient to groups of covariates.

II.1 UPPER SECONDARY REGRESSION RESULTS

We begin by estimating equation (2) for enrolment in practical STEM subjects at the LC level. The regression includes grouped controls capturing: STEM-readiness, comparative advantage, preferences and tastes, school and peer factors, and background characteristics. Appendix Table A.5 reports coefficient estimates for the full specification.

The results indicate that enrolment in practical STEM subjects at LC level is strongly associated with prior curricular exposure. In particular, students who studied practical subjects at lower secondary level (material technology, metalwork, technical graphics, and technology) are significantly more likely to enrol in practical STEM subjects at upper secondary. In addition, students who took a greater number of higher-level subjects during the Junior Cycle are more likely to participate in practical STEM later. These findings point to the importance of early curricular pathways. Students who select into practical or technically oriented subjects earlier in secondary school are substantially more likely to continue along a similar trajectory at LC level. This suggests that participation in practical STEM at upper secondary is largely the continuation of pathways established during lower secondary education. By the time students make their LC subject choices, a considerable portion of the divergence in participation has already emerged through earlier subject selection and exposure.

In addition to earlier subject choices, several additional factors are associated with practical STEM participation. Students whose favourite JC subject is a practical subject are significantly more likely to enrol in practical STEM, as are those who report occupational aspirations in transport or agriculture-related fields. School-level subject availability also matters: attending a school that offers agricultural science or biology is positively associated with practical STEM enrolment. In contrast, students who dislike maths are less likely to take practical STEM subjects than those who always like maths. Similarly, students whose primary caregiver was born outside Ireland are less likely to participate. Finally, higher reading scores at age nine are negatively associated with practical STEM enrolment, consistent with a comparative advantage mechanism whereby students with stronger verbal skills are less likely to specialise in practical or technical subjects.

II.2 UPPER SECONDARY GENDER GAP DECOMPOSITION

We now turn to gender differences in practical STEM enrolment in [Table 3](#). The unconditional gender gap is large: 45.96% of male students take at least one practical STEM subject, compared to 15.63% of female students, resulting in a gap of 30.33 percentage points. This raw gap corresponds to the coefficient on the female indicator in a baseline specification that includes only gender. Conditioning on the full set of covariates reduces the estimated female coefficient to 7.08 percentage points. Observable differences captured by the covariate set therefore explain a substantial proportion of the unconditional disparity.

To understand which mechanisms account for the reduction in the female coefficient from the baseline to the full model, we apply the [Gelbach \(2016\)](#) decomposition. Panel B of Table 3 reports the results of the decomposition, showing the extent to which each group of covariates accounts for the change.

The dominant contribution comes from STEM-readiness, captured by JC subject choices and related indicators: this group accounts for 17.41 percentage points of the reduction in the gender coefficient ($17.41/30.33 = 57.4\%$ of the total gap). This implies that a large share of the gender gap observed at the end of secondary school reflects earlier differences in curricular pathways established by lower secondary subject choices, typically made around age 13. These results reinforce a pipeline interpretation, mirroring the well-established finding that access to and enrolment in STEM in higher education depend on preparation accumulated during secondary school. Our contribution is to show that the same logic applies one step earlier: the STEM-readiness that matters for higher education is itself shaped by choices made at the start of secondary school. Many students, particularly female students, may have already limited their future STEM opportunities well before reaching the senior cycle.

Preferences and tastes explain an additional 5.84 percentage points (19.3% of the total gap). Comparative advantage (skills), school and peer factors, and background characteristics contribute relatively little to the change in the female coefficient. This suggests that gender differences in practical STEM participation at the LC level are not primarily a story of girls lacking the measured skills needed to succeed, nor a story driven by socioeconomic background. Rather, gender differences at the end of secondary school are largely explained by earlier curricular sorting and persistence: once students have entered different subject pathways, subsequent choices reflect that accumulated exposure.

Table 3: LC Practical STEM — Gender gap and Gelbach decomposition

<i>Panel A: OLS estimates</i>		
Baseline - Female (1)	30.33***	(0.02)
Full Model - Female (2)	7.08***	(0.03)
Observations	3,888	
<i>Panel B: Gelbach decomposition of the change in the gender gap</i>		
Total change: (1)–(2)	23.25***	(0.03)
<i>Contribution to change:</i>		
STEM-readiness	17.41***	(0.02)
Comparative advantage	0.04	(0.00)
Peer effects	0.00	(0.02)
Preferences	5.84***	(0.01)
Background	-0.04	(0.00)

Note: Panel A reports the coefficient on the female indicator from (i) a baseline specification including only gender and (ii) a full specification including controls for STEM-readiness, comparative advantage, preferences/tastes, school/peer factors, and background characteristics. Panel B reports a Gelbach (2016) decomposition of the difference in the Female coefficient between the baseline and full specification, allocating the total change across pre-defined covariate groups. The dependent variable is an indicator equal to one if the student takes at least one practical STEM subject at Leaving Certificate level and zero otherwise. Robust standard errors clustered by school in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

III. RESULTS - LOWER SECONDARY

Given that enrolment in practical STEM subjects at the lower secondary level is the strongest predictor of enrolment in similar subjects at the upper secondary level, a natural next step is to examine the determinants of these earlier choices. We therefore model students' decisions to enrol in practical STEM subjects at the lower secondary stage. Our analysis relies on explanatory variables measured in Wave 1, when students were aged nine and had not yet entered secondary school or made subject choices. These variables capture a broad set of early-life characteristics, including measures of cognitive and non-cognitive skills, as well as proxies for students' interests, aspirations, and preferences, allowing us to address theories of comparative advantage and individual preferences. As the explanatory variables are measured prior to secondary school entry, we do not include secondary school characteristics in this specification. Our aim is to isolate factors that shape subject choices before students select into particular schools or are exposed to secondary school environments. We therefore examine the role of school environments separately in subsequent analyses and robustness checks. Similarly, we do not include a measure of STEM-readiness in the lower secondary decomposition. While STEM-readiness is an important predictor of STEM participation in upper secondary and higher education, there is essentially no variation in STEM-readiness prior to secondary school, as all students follow the same national primary school curriculum. STEM-readiness can therefore be discounted

as a potential driver of the gender gap at lower secondary.

As in the upper secondary analysis, we proceed in two steps. First, we present regression results describing the correlates of enrolment in practical STEM subjects at the JC level. Second, we quantify the gender gap at this stage, show how it changes when conditioning on observables measured at age 9, and apply a Gelbach decomposition to attribute the change in the gender coefficient between the baseline and full specifications to groups of covariates.

III.1 LOWER SECONDARY - SUMMARY STATISTICS

There are 4 practical STEM subjects which students in our sample could take at the lower secondary level: Material Technology (MT), Technical Graphics (TG), Metalwork, and Technology. The decomposition results from the previous section suggest that enrolment in practical STEM subjects at the lower secondary level is a key driver of participation in these subjects at the upper secondary level, which in turn influences college STEM enrolment. Each subject can be considered a “feeder” subject for a corresponding upper-secondary subject. 79% of students in our sample who take Construction at Leaving Cert had taken Material Technology at Junior Cert, 76% who take DCG at Leaving Cert had taken TG at Junior Cert, 68% who take Engineering had taken Metalwork at Junior Cert, and 54% who take Technology at Leaving Cert had taken Technology at Junior Cert.⁶ [Table 4](#) shows the proportion of each gender taking each subject. Enrolment rates for each STEM subject at the national level are provided to show that the sample is representative of individuals taking the JC in Ireland during the sample period. See [Appendix Section 7.2](#) for these statistics.

To ensure consistency between our lower secondary and upper secondary analyses, we define the composite outcome variable *Practical JC* as 1 if a student studies at least one of the four practical subjects and 0 otherwise. The gender gap at this stage is even larger than at LC level: 72.55% of male students have a value of *Practical JC* = 1, compared to 13.78% of female students, a difference of 58.77 percentage points.

Table 4: JC Practical Enrolment by Gender

Subject	Male (%)	Female (%)	Total (%)
<i>Practical STEM Subjects</i>			
Material Technology	41.57	5.54	24.01
Technical Graphics	33.25	5.93	19.93
Metalwork	18.12	1.83	10.18
Technology	10.97	1.96	6.58
<i>Practical STEM</i> (1+ practical subject)	72.55	13.78	43.91
Total Observations	1,906	1,982	3,888

Notes: Percentages report weighted enrolment rates by gender for Junior Certificate (JC) Practical STEM subjects; for example, 41.57% of male students, 5.54% of female students, and 24.01% of all students in the sample study material technology for JC. The composite row (“Practical STEM”) indicates whether a student is enrolled in at least one of these subjects.

III.2 LOWER SECONDARY REGRESSION RESULTS

The main specification regresses Practical JC on a rich set of covariates measured at age 9. Appendix Table A.6 reports the estimates from the fully specified model of this education stage. Gender itself is by far the most influential factor, with the results indicating that gender is the dominant predictor of practical STEM enrolment at lower secondary level. The majority of other variables are statistically insignificant, suggesting that early-life characteristics explain relatively little of the variation in practical STEM participation at this stage.

There are, however, exceptions: students with a lower Drumcondra reading score at age 9 are more likely to enrol in practical STEM subjects, consistent with a comparative advantage mechanism in which students with relatively weaker verbal skills are more likely to specialise in practical or technical subjects. There is also weak evidence that students who report never liking school are more likely to enrol in practical STEM subjects, although this effect is only marginally statistically significant. Students who aspire to a career in a technical or trade/manual sector are more likely to enrol in practical STEM subjects, highlighting the role of early career preferences in shaping subject choices, even at the lower secondary level. Finally, students whose parents do not have a college education are more likely to enrol in practical subjects, suggesting that socioeconomic background is associated with selection into different curricular pathways.

III.3 LOWER SECONDARY GENDER GAP DECOMPOSITION

The unconditional gender gap in *Practical JC* is 58.77 percentage points, corresponding to the female coefficient in a baseline model that includes only a gender indicator. Conditioning on the full set of age-9

covariates reduces the female coefficient only slightly, from 58.77 to 56.98, a 1.79 percentage-point reduction. Thus, only 3% of the lower secondary gender gap is explained by the observable characteristics available in our data prior to secondary school entry.

Table 5 reports the results of the Gelbach decomposition. Preferences and tastes account for the largest share of the explained component, but their contribution is modest in absolute terms, reducing the female coefficient by 2.21 percentage points. Comparative advantage and background characteristics contribute negligibly, with effects close to zero. The vast majority of the disparity remains unexplained.

The lower secondary results imply that the practical STEM gender gap cannot be accounted for by a wide range of differences observable in late primary school. The GUI data contain unusually rich measures of cognitive skills, non-cognitive traits, academic self-concept, early interests, occupational aspirations, family background, and later school characteristics. Yet, despite this breadth of information, conditioning on these measures explains only a trivial share of the JC gap. In this context, the results allow us to reject several common explanations for gender differences in STEM-related choices. The gap does not appear to be driven primarily by measured differences in preferences, family background, comparative advantage, or STEM-readiness

One possibility is that the relevant drivers are forms of gendered meaning or socialisation that are only weakly captured by standard survey measures at this age: for example, beliefs about which subjects are “for boys” or “for girls”, anticipated social costs of choosing gender-atypical subjects, or differential encouragement from parents, teachers, or peers. These mechanisms are consistent with our results precisely because they are not necessarily measured by the rich set of explanatory variables included in our models. In other words, the unexplained component should not be interpreted as an absence of mechanism, but rather as evidence that the mechanisms most commonly emphasised in the literature do not account for this gap in our setting. An alternative interpretation is that the divergence may emerge primarily between ages 9 and 13, through processes that occur during late primary school and the transition into adolescence.

Either interpretation is inconsistent with a simple story in which girls at age 9 already express systematically different aspirations that map cleanly into later practical STEM participation. Our preference and aspiration measures, including early reported interests and occupational aspirations, contribute little to explaining the gap. If gendered attitudes are a key driver, they appear not to be captured by these early self-reports.

Considering each theoretical explanation individually provides further clarity. First, as previously mentioned, STEM-readiness is not a relevant mechanism at the lower secondary level, so it can be ruled out as a

potential explanation for the gender gap. Second, the comparative advantage theory, as captured by cognitive assessments, explains only a negligible proportion of the gender gap. This indicates that skill differences observable at age 9 play minimal roles in determining subsequent practical STEM subject choices.

Third, preference-based theories, proxied by students' early reported feelings towards maths, reading and school in general, as well as their occupational aspirations, contribute slightly more, but still explain only a small portion of the gender disparity (3.76% of the total gap). While our data allow for a broad approximation of early preferences, capturing every relevant dimension of student interests and aspirations at this age is nearly impossible. As such, we cannot definitively rule out that underlying differences in preferences, potentially shaped by early socialisation or gender norms, are driving the observed gap. This finding suggests that the gender gap is not driven by differential educational or career aspirations, or differences in general interests or enjoyment of core subjects, but could be driven by attitudes or stereotypes that we do not directly measure.

Fourth, background characteristics, including socioeconomic indicators and family conditions, appear to have no meaningful explanatory power for the gender gap. This aligns with expectations, as background variables, while potentially influential on overall enrolment decisions, are the same on average for male and female students. Lastly, due to the absence of school-level factors in the lower secondary analysis, a deeper examination of school and peer effects at later stages of education is warranted. We address this in the next section.

Overall, the overwhelming dominance of the residual gender component in this early-stage decomposition underscores the importance of exploring intermediate educational stages more closely, with particular attention to socialisation, gender norms and teacher interactions. Targeted policy interventions and mentorship programmes designed to counteract these factors prior to secondary school may prove vital for mitigating the substantial gender gap observed in practical STEM subjects at both lower and upper secondary levels, and the subsequent gender gaps in college.

IV. ROBUSTNESS TESTING

IV.1 MACHINE LEARNING

The preceding results show that while a large share of the gender gap in LC practical STEM enrolment can be traced back to JC subject choices, the (even larger) gender gap in JC practical STEM enrolment is almost entirely unexplained by the rich set of characteristics we observe at age 9. One concern is that this finding

Table 5: JC Practical STEM — Gender gap and Gelbach decomposition

<i>Panel A: OLS estimates</i>		
Baseline - Female (1)	58.77***	(0.02)
Full Model - Female (2)	56.98***	(0.02)
Observations	3,888	
<i>Panel B: Gelbach decomposition of the change in the gender gap</i>		
Total change: (1)–(2)	1.79*	(0.01)
<i>Contribution to change:</i>		
Comparative advantage	-0.21	(0.00)
Preferences	2.21**	(0.01)
Background	-0.19	(0.00)

Note: Panel A reports the coefficient on the female indicator from (i) a baseline specification including only gender and (ii) a full specification including controls for comparative advantage, preferences/tastes, and background characteristics. Panel B reports a Gelbach (2016) decomposition of the difference in the Female coefficient between the baseline and full specification, allocating the total change across pre-defined covariate groups. N=3,888. The dependent variable is an indicator equal to one if the student takes at least one practical STEM subject at Junior Certificate level and zero otherwise. Robust standard errors clustered by school in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

could be an artefact of our modelling choices: the linear probability model imposes strong functional-form restrictions, and our decomposition relies on a pre-specified set of covariates grouped into broad mechanisms. To assess whether the unexplained component is driven by these parametric assumptions or by variable selection, we complement the regression-based analysis with a flexible machine-learning approach that allows for rich non-linearities and interactions, and lets the data determine which variables are most predictive.

Specifically, we fit gradient-boosted decision tree (XGBoost) models to predict practical STEM participation at both LC and JC stages, using an expanded set of available covariates.⁷ Our main specification includes a set of covariates linked to theoretical mechanisms, which have been highlighted in the literature as potential explanations for gender gaps in STEM-based choices. This allows for an interpretable assessment of their contribution to the gender gap while avoiding overly broad controls in the main analysis. The machine learning analysis draws on a broader set of variables from the GUI data to capture additional patterns. We then decompose the predicted gender gap using SHAP (Shapley Additive Explanations) values, which attribute each individual prediction to contributions from each feature. Aggregating SHAP values across individuals allows us to express the model-implied gender gap as the sum of contributions from our mechanism groups (STEM-readiness, comparative advantage, preferences, school factors, background), with the remaining component captured by the SHAP contribution of the gender indicator itself (i.e. the part not explained by other observed characteristics in the model). We provide full details on model tuning, train/test splits, and variable construction in Appendix [Section 7.4](#) and [Section 7.5](#).

The machine-learning results in [Table 6](#) closely mirror the regression-based conclusions. At the upper-secondary stage, the model predicts a gender gap in LC practical STEM participation similar to the observed gap, and the decomposition attributes a substantial share of this gap to differences in JC subject choices (STEM-readiness), with smaller roles for other covariate groups. At the lower secondary stage, the model again predicts a large gender gap in JC practical STEM participation, but the decomposition assigns only a minimal share of the gap to observed characteristics measured at age 9, with the vast majority attributed to the residual direct contribution of gender. The persistence of this pattern in a flexible, high-dimensional setting suggests that the large unexplained gender gap at the lower secondary level is unlikely to be driven by restrictive functional-form assumptions or by an overly narrow choice of explanatory variables. Moreover, the fact that moving to a substantially richer, more flexible set of predictors, including a broader set of child attitudes, behavioural measures, school experiences, and family background indicators (see Appendix Table A.4), leaves the explained share essentially unchanged suggests that the result is unlikely to be driven by age-9 characteristics available in the GUI survey but excluded from our baseline model. Instead, the evidence points to forces that emerge after age 9, during late primary school or the transition into adolescence, or to mechanisms that are not well captured by standard survey measures at that age.

Table 6: SHAP Decompositions of Gender Gaps in Practical STEM (LC and JC)

	LC Practical STEM SHAP	JC Practical STEM SHAP
Observed gap (test data)	28.87	58.62
Predicted gap (test data)	31.15	50.72
<i>Contribution to predicted gap:</i>		
STEM-readiness/Junior Cert	14.33	—
Peer effects/School factors	1.94	0.61
Preferences/Tastes	5.45	2.82
Comparative advantage/Skills	1.00	0.63
Background	0.17	0.20
Gender (residual)	8.26	46.46

Notes: Results from XGBoost models predicting practical STEM participation at LC and JC levels. Models are estimated using a 70% training / 30% test split. “Observed gap” and “predicted gap” refer to weighted gender differences in mean outcomes in the test data, measured in percentage points. Contribution rows report each covariate group’s share of the predicted gender gap based on SHAP (Shapley Additive Explanations) values aggregated over the test sample. “Gender (residual)” captures the residual contribution of the gender indicator not explained by other observed variables. “—” indicates that the covariate group is not included in the JC specification. The reported sample corresponds to the held-out test set.

IV.2 THE ROLE OF SCHOOL

Ireland is a particularly interesting case study when examining gender disparities in secondary school, as it has an unusually high prevalence of single-sex schools. If male-only and female-only schools tend to offer different subject ranges to their students, it is possible that gender differences in subject enrolment reflect school-level sorting rather than individual preferences or characteristics. As such, a potential concern is that the subject choices analysed in the previous section may be heavily influenced by a student’s (or parent’s) choice of secondary school. In this section, we examine whether our results are primarily driven by differences between schools or whether the gender gaps reflect deeper patterns rooted in students’ decisions during lower secondary education.

We begin by examining variation in subject offerings across secondary schools of different gender compositions (boys-only, girls-only, and mixed) in [Figure A.1](#). Due to data restrictions from the data provider, we can not show enrolment rates for Construction and Engineering in Girls-only schools in our sample at the LC level. Design and Communication Graphics and Technology are rare in girls-only schools, though they are uncommon in male schools as well. The same pattern is observed at the Junior Cert level; girls-only schools are less likely than boys-only schools to offer all of Material Technology, Technical Graphics, Metalwork and Technology, although all of these subjects are more commonly offered in mixed schools than in boys-only schools. As male and female students are systematically sorted into different schools that tend to offer different subject sets, these disparities could be driving our results. Single-sex schools account for a large proportion of our sample, with roughly 23% of students attending female-only schools and a further 23% attending male-only schools. National statistics are provided for comparison in [Appendix Table A.3](#).

If school subject offerings were the primary driver of gender disparities in practical STEM enrolment, we would expect these gaps to diminish in coeducational settings where male and female students have identical subject choice sets. However, as shown in [Table 7](#), the reverse often holds. Across both the Junior and Leaving Certificate cycles, gender gaps in practical STEM participation are larger within mixed schools, suggesting that even when male and female students attend the same institution and have equivalent subject options, female students are far less likely to choose practical courses. The only exception to this finding is Construction which shows a slightly larger gender gap within single sex schools.⁸

This finding contradicts the notion that single-sex schools are the primary source of gender imbalances in practical STEM uptake. Instead, it indicates that gender differences in subject preferences and choices persist and potentially intensify in environments with equal access to the full subject choice set. Such a finding is especially noteworthy given that mixed schools represent the single largest category in our sample.

Table 7: Gender Gaps in Subject by School Type

Subject	Gender Gap - Single Sex (%)	Gender Gap - Mixed (%)
Leaving Cert outcomes		
LC Practical STEM	25.49	33.58
Leaving Cert subjects		
Construction Studies	13.42	25.80
Engineering	4.10	20.88
Design and Communication Graphics	12.12	11.72
Agricultural Science	4.47	4.80
Technology	3.78	5.19
Junior Cert outcomes		
JC Practical STEM	52.90	62.45
Junior Cert subjects		
Material Technology	26.01	43.50
Metalwork	4.58	25.46
Technical Graphics	26.74	27.38
Technology	7.02	10.62
Observations	1,900	1,988

Note: Gender gaps are calculated as the difference in weighted mean enrolment rates between male and female students (male minus female), expressed in percentage points. Positive values indicate higher enrolment among male students. “Single-sex schools” combine boys-only and girls-only schools; “mixed schools” include coeducational schools only. Estimates are computed separately within each subsample. The full sample includes 3,888 students.

The patterns observed in our data challenge the idea that school choice is the primary driver of gender differences in STEM trajectories. Although some single-sex schools offer a narrower range of practical STEM subjects, this cannot account for the substantial gender gaps we observe across the broader student population. In fact, the persistence of these gaps in mixed-sex schools, where male and female students face the same set of subject options, suggests that differences in individual choices rather than school-level constraints play a more important role in explaining gender differences in STEM participation. This implies that the patterns documented in the previous section are primarily driven by JC subject choices rather than by differences arising from school choice.

Nevertheless, concerns may remain that male and female students sort into schools that differ either in the subjects they offer or in some unobservable characteristics which affect subject choice. To address this directly, we repeat our analysis using more restrictive samples in [Table 8](#). We limit the sample to mixed-sex schools and re-estimate our practical STEM participation models. In addition, we include school fixed effects in these specifications. This approach allows us to examine the gender gap within schools by comparing male and female students who face the same institutional environment, including identical subject availability, teacher resources, and peer composition. By relying on within-school variation, this specification removes

cross-school differences that might otherwise influence subject choice.

Table 8: Practical STEM — Gender gap and Gelbach decomposition Mixed Schools (Fixed Effects)

	LC	JC
<i>Panel A: OLS estimates</i>		
Baseline with School FEs - Female (1)	34.46*** (0.03)	63.31*** (0.03)
Full Model with School FEs - Female (2)	7.61** (0.04)	59.54*** (0.03)
Observations	1,988	1,988
<i>Panel B: Gelbach decomposition of the change in the gender gap</i>		
Total change: (1)–(2)	26.85*** (0.03)	3.77** (0.02)
<i>Contribution to change:</i>		
STEM-readiness	22.13*** (0.03)	
Comparative advantage	-1.05 (0.01)	0.36 (0.00)
Peer effects	0.00 (0.00)	
Preferences	6.07*** (0.02)	3.44** (0.02)
Background	-0.30 (0.01)	-0.03 (0.01)

Note: Results are estimated on a restricted sample of students attending mixed-sex schools only. All specifications include school fixed effects, which absorb all time-invariant school-level characteristics, including subject availability, school type, and peer composition. Panel A reports the coefficient on the female indicator from (i) a baseline specification including only gender and school fixed effects and (ii) a full specification including controls. Panel B reports a Gelbach (2016) decomposition of the difference in the Female coefficient between the baseline and full specification, allocating the total change across covariate groups. Blank cells indicate covariate groups not included in the JC specification. Robust standard errors clustered at the school level are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Our results remain qualitatively unchanged across specifications. Even when limiting the sample to mixed-sex schools and including school fixed effects, the estimated gender gaps are similar in magnitude to those in the main analysis. The Gelbach decomposition still highlights the role of prior subject choices in explaining gender differences in LC practical STEM participation. As shown in Table 8, over 64% of the explained LC gender gap stems from earlier subject choices at the JC stage.

In contrast, the JC gender gap decomposition shows most of the difference remains unexplained by characteristics observed at age nine. In the "JC" column of Table 8, we see that approximately 94% of the gender gap in JC practical STEM enrolment cannot be explained by early-life background, ability, or preference indicators. Taken together, these results suggest that the gender gap is not driven by school-level

factors such as subject availability or school selection. When we restrict the sample to mixed-sex schools and exploit within-school variation using school fixed effects, the estimated gaps and decomposition results remain largely unchanged, reinforcing the conclusion that the divergence arises early in students’ educational trajectories and is not explained by the observed characteristics we consider.

V. CONCLUSION

We examine gender disparities in STEM subject enrolment in upper secondary school, a key predictor of STEM participation in higher education. While STEM is often treated as a single category in the literature, it encompasses a diverse set of disciplines, and the gender gaps are considerably larger in practical subjects than in traditional science and maths. We identify the key predictors of enrolment in practical STEM subjects during secondary education using rich data on individuals’ abilities, interests, preferences, self-perceived ability, prior experiences, background characteristics, and school-level factors. We find that subject choices made at the lower secondary level, typically around age 13, are the strongest predictor of practical STEM participation at the upper secondary level. These early choices account for 50–60% of the observed gender gap in practical STEM enrolment at upper secondary school.

At the lower secondary level, the rich set of variables observed at age 9 explains only a small share of the gender gap in practical STEM enrolment: over 96% of the gap remains unexplained. This is a notable result given the breadth of the GUI data. In this setting, the results allow us to rule out several common explanations for gender differences in STEM-related choices: the early practical STEM gap is not primarily explained by measured differences in ability, broad early preferences, school-level factors, or family background. This suggests that factors beyond early observable characteristics drive divergent curricular choices between boys and girls by early adolescence, a divergence that has materialised by age 13.

Our findings are also robust to a richer and more flexible modelling strategy and to within-school comparisons. Expanding the covariate set in the machine-learning exercise leaves the explained share of the JC gap essentially unchanged, suggesting that the result is not an artefact of our baseline variable selection or linear specification. Likewise, the gender gap remains large in mixed-sex schools and persists when we compare students within the same school using fixed effects. These results indicate that the early divergence in practical STEM subject choice is not driven by differences in school environments, subject availability, or sorting across schools.

The findings of this paper offer clear guidance for policy design. If efforts to reduce gender disparities in practical STEM fields are to succeed, timing is critical. Gender differences in enrolment emerge as soon

as subject choice becomes available around age 13, despite similar academic profiles, measured preferences, and background characteristics at age nine. This suggests that interventions should be introduced before subject choices are made, in the years preceding lower secondary school. The evidence points away from standard skill-based and school-constraint explanations and toward mechanisms that are less well captured by conventional survey measures, such as gender norms, stereotypes, and differential social encouragement around practical subjects. Policies should therefore focus on addressing preferences and attitudes through encouragement, information provision, and efforts to challenge gender stereotypes, rather than initiatives aimed primarily at skill development, academic streaming, or structural changes to school types.

In conclusion, although this study is descriptive rather than causal, it highlights a critical window for intervention and points to concrete policy levers. The persistence of large, unexplained gender gaps in practical STEM participation, even among students with comparable early academic profiles, underscores the need to directly shape perceptions, attitudes, and aspirations before students make subject choices. Secondary school subject selection marks a key juncture in the STEM pipeline, and early, targeted engagement is likely essential to reducing entrenched gender disparities and strengthening the future STEM workforce.

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Permission to reproduce material from other sources: No copyrighted material requiring permission has been

used in this study.

NOTES

¹Examples of government policies to promote STEM participation among women include the USA’s INSPIRE Women Act (2017) and Australia’s Advancing Women in STEM Strategy.

²Similar to (Doris, O’Neill and Sweetman, 2022), we convert the grades for the JC into a numerical value, with the maximum score a student can get in any subject being 12 and the minimum being 1.

³The Strengths and Difficulties Questionnaire (SDQ) serves as a behavioural screening instrument. The primary caregiver (over 99% of PCGs are recorded as the mother of the child) completes the questionnaire, which assesses 25 attributes relating to their child and covers topics such as emotional, conduct, and prosocial behaviour. A total score is calculated from the questionnaire, with higher scores indicating greater behavioural and emotional difficulties.

⁴See <https://www.curriculumonline.ie/> for further information on each subject’s curriculum.

⁵In addition to the aggregate practical STEM outcomes, we also examine gender gaps separately for individual subjects. The corresponding results are reported in Appendix Tables A.8 and A.9.

⁶Further information on each subject’s curriculum can be found at <https://www.gov.ie/en/department-of-education/collections/junior-certificate-syllabuses-and-prescribed-material/>.

⁷In addition to the covariates used in the decomposition analysis, the machine learning models include an additional 66 variables capturing student skills, preferences, background characteristics, and school factors. These extra variables are reported in Appendix Table A.4.

⁸Consistent with our findings, Delaney and Devereux (2019) find an approximately four percentage-point larger overall STEM gender gap in mixed-sex schools. They also note that factors beyond subject availability, including peer effects and gender stereotypes, may contribute to these differences in subject choice. These findings also align with those of Schneeweis and Zweimüller (2012), who find that exposure to peers of the same gender decreases the likelihood of female students sorting into traditionally female-dominated fields.

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VI. APPENDIX

VI.1 SUMMARY STATISTICS AND INFORMATION

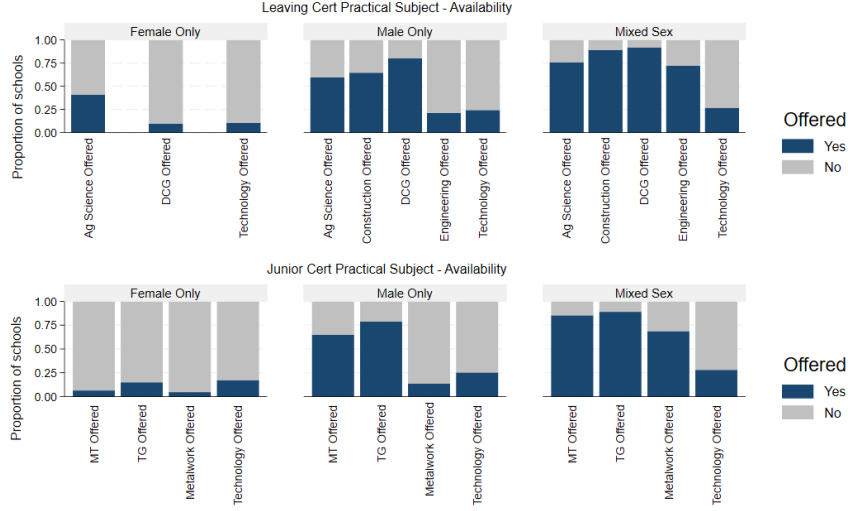


Figure A.1: Subject Availability by School Sex - Upper and Lower Secondary

Table A.1: Summary Statistics of School Characteristics

	Total (%)	Pre (%)	Post (%)	Weighted - Post (%)
<i>School Gender</i>				
Single Sex - Boys	16.01	22.87	23.34	23.23
Single Sex - Girls	20.65	24.98	25.58	25.64
Mixed Sex	63.34	52.15	51.08	51.13

Note: Distribution of school gender composition in the sample and in national statistics. “Total” refers to national figures for Irish secondary schools. “Pre” refers to the full GUI sample; “Post” refers to the sample after restrictions; “Weighted-Post” applies survey weights to the restricted sample. Total figures are taken from national statistics for the school years 2015/2016 and 2016/2017. For reference see secondary school information via <https://data.cso.ie/>

Table A.2: Observations lost per sample restriction imposed

<i>Variable</i>	<i>Sample</i>	<i>Observations dropped</i>	<i>Sample left</i>
<i>Wave 1</i>			
PCG being the Mother	6,039	71	5,968
Books in home	5,968	2	5,966
PCG Foreign Born	5,966	4	5,962
Like Maths	5,962	21	5,941
Drumcondra Maths	5,941	58	5,883
Drumcondra Reading	5,883	69	5,814
<i>Wave 2</i>			
Academic Self Concept	5,814	69	5,745
Conscientiousness	5,745	6	5,739
Practical favourite subject	5,739	182	5,577
Own expectations	5,577	83	5,494
Maths Difficult	5,494	6	5,488
English Difficult	5,488	3	5,485
<i>Wave 3</i>			
Doing English	5,485	41	5,444
Doing Higher English	5,444	2	5,442
Doing Maths	5,442	6	5,436
Transition Year	5,436	65	5,371
Number of higher level subjects	5,371	13	5,358
English JC Grade	5,358	39	5,319
Maths JC Grade	5,319	2	5,317
Principal - School Gender	5,317	1,385	3,932
Principal - School Type	3,932	37	3,895
Principal - DEIS Status	3,895	7	3,888

Note: The table documents the sequential sample restrictions applied to construct the dataset. The initial balanced panel consists of 6,039 students participating in the first 3 waves of GUI. Each row shows the number of observations dropped due to missing data for a given variable and the remaining sample size after the restriction. A large number of principals did not answer the questionnaire relating to school characteristics and subject offerings etc. Hence, this is where we see the largest loss of observations. PCG = Primary Caregiver.

VI.2 COMPARISON OF ENROLMENT RATES

Table A.3: Subject Enrolment Rates for Practical Subjects - LC & JC

Subject	Gender	2015	2016	2017	2018	Total (Average) %	Pre %	Post %	Weighted - Post %
<i>LC - Total Enrolment</i>									
Total		55,006	55,684	55,731	54,396	-	-	-	-
	Female	27,815	27,790	27,886	27,576	-	-	-	-
	Male	27,192	27,894	27,845	26,820	-	-	-	-
<i>LC - Practical STEM</i>									
Construction Studies		8,269	8,553	8,750	8,248	15.32	11.14	10.23	12.28
	Female	676	763	859	902	1.45	1.06	0.66	0.90
	Male	7,593	7,790	7,891	7,346	13.87	10.08	9.57	11.38
Agricultural Science		7,672	7,893	7,660	7,778	14.04	11.33	10.70	11.90
	Female	2,920	3,048	3,093	3,152	5.53	5.19	4.65	4.79
	Male	4,752	4,845	4,567	4,626	8.51	6.14	6.05	7.11
DCG		5,362	5,523	5,575	5,393	9.90	9.62	9.43	9.72
	Female	649	661	729	760	1.27	1.47	1.61	1.80
	Male	4,713	4,862	4,846	4,633	8.63	8.15	7.82	7.92
Engineering		5,376	5,379	5,275	5,254	9.64	6.68	6.18	7.64
	Female	324	298	315	355	0.59	0.40	0.39	0.39
	Male	5,052	5,081	4,960	4,899	9.05	6.28	5.79	7.25
Technology		1,328	1,415	1,527	1,534	2.63	2.85	2.96	3.38
	Female	233	231	271	247	0.40	0.59	0.52	0.54
	Male	1,095	1,184	1,256	1,287	2.23	2.26	2.44	2.84
	Gender	2012	2013	2014	2015				
<i>JC - Total Enrolment</i>									
Total		58,798	59,822	60,327	59,522	-	-	-	-
	Female	28,891	29,269	29,277	29,213	-	-	-	-
	Male	29,907	30,553	31,050	30,309	-	-	-	-
<i>JC - Practical STEM</i>									
Material Technology		15,775	16,163	16,464	16,145	27.08	21.44	20.07	24.01
	Female	1,964	2,139	2,145	2,500	3.67	2.84	2.18	2.80
	Male	13,811	14,024	14,319	13,645	23.41	18.60	17.89	21.21
Technical Graphics		12,004	12,564	12,349	11,701	20.39	20.03	19.50	19.93
	Female	1,705	1,805	1,758	1,746	2.94	2.97	3.04	3.01
	Male	10,299	10,759	9,871	9,955	17.45	17.06	16.46	16.92
Metalwork		7,739	7,843	7,880	7,984	13.19	9.21	8.34	10.18
	Female	717	623	742	797	1.21	0.86	0.80	0.90
	Male	7,022	7,220	7,138	7,187	11.98	8.35	7.54	9.28
Technology		3,026	2,957	3,223	3,258	5.23	5.22	5.76	6.58
	Female	618	525	531	612	0.96	1.02	0.88	1.01
	Male	2,408	2,432	2,692	2,646	4.27	4.20	4.88	5.57

Note: ‘Total’ column is the national average. ‘Pre’ column is the unweighted sample before data restrictions. ‘Post’ column is the unweighted sample after data restrictions. ‘Weighted - Post’ column is the weighted sample after data restrictions. National figures are from the State Examinations Statistics (SEC) <https://www.examinations.ie/statistics/>

VI.3 FULL LIST OF VARIABLES

Table A.4: Summary of Variables

Variable	Group	Age	LC/JC	Description	Mean
<i>Main Analysis - Variables</i>					
Gender	Gender	9	Both	=1 if student is female	0.49
Studied Foreign Language for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC French, German or Spanish	0.90
Studied Arts for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC Art	0.33
Studied Music for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC Music	0.20
Studied Home Economics for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC Home Economics	0.33
Studied Material Technology for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC Material Technology	0.24
Studied Metalwork for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC Metalwork	0.10
Studied Technical Graphics for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC TG	0.20
Studied Business Studies for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC Business Studies	0.59
Studied Technology for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC Technology	0.07
Studied Religion for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC Religion	0.50
Studied Higher-Level English for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC English at Higher Level	0.83
Studied Higher-Level Maths for JC	STEM-readiness/JC	17/18	LC	=1 if student took JC Maths at Higher Level	0.65
Did Transition Year	STEM-readiness/JC	17/18	LC	=1 if student took Transition Year	0.71
Drumcondra Reading Score	Comp. Adv/Skills	9	Both	Standardised reading test	0.12
Drumcondra Numerical Score	Comp. Adv/Skills	9	Both	Standardised math/numerical	-0.65
Academic Self-Concept	Comp. Adv/Skills	13	LC	Academic self-concept scale	12.10
Conscientiousness Score	Comp. Adv/Skills	13	LC	Conscientiousness measure	4.23
Strengths & Difficulties Questionnaire (SDQ) Score	Comp. Adv/Skills	13	Both	Non-cognitive measure	6.58
Number of Higher-Level Subjects taken for JC	Comp. Adv/Skills	17/18	LC	Number of JC subjects at higher level, 1–12	8.21
JC English Grade	Comp. Adv/Skills	17/18	LC	Letter grade converted	9.87
JC Maths Grade	Comp. Adv/Skills	17/18	LC	Letter grade converted	9.30
School Gender 1	School/Peer Effects	17/18	LC	=1 if Female-only school	0.23
School Gender 2	School/Peer Effects	17/18	LC	=1 if Male-only school	0.26
School Gender 3	School/Peer Effects	17/18	LC	=1 if Mixed-sex school	0.51
School Type 1	School/Peer Effects	17/18	LC	=1 if Fee-paying school	0.11

School Type 2	School/Peer Effects	17/18	LC	=1 if Non-fee-paying secondary	0.49
School Type 3	School/Peer Effects	17/18	LC	=1 if Vocational school	0.12
School Type 4	School/Peer Effects	17/18	LC	=1 if Community/comprehensive	0.27
Attended DEIS School	School/Peer Effects	17/18	LC	=1 if DEIS school	0.15
French Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC French	0.99
German Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC German	0.67
Spanish Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Spanish	0.42
Arts Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Arts	0.95
Applied Maths Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Applied Maths	0.45
Physics Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Physics	0.92
Chemistry Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Chemistry	0.94
Agricultural Science Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Agricultural Science	0.66
Biology Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Biology	0.99
Engineering Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Engineering	0.47
Construction Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Construction	0.65
Accounting Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Accounting	0.88
Business Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Business	0.97
Economics Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Economics	0.61
Technology Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Technology	0.22
Music Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Music	0.88
Home Economics Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Home Economics	0.77
Project Maths Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Project Maths	0.73
Religious Studies Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Religious Studies	0.51
Link Modules Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Link Modules	0.63
DCG Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC DCG	0.71
Other Languages Offered in School	School/Peer Effects	17/18	LC	=1 if school offers other LC languages	0.08
YP Likes Maths	Preferences/Tastes	9	Both	1=Always, 2=Sometimes, 3=Never	1.61
YP Likes Reading	Preferences/Tastes	9	Both	1=Always, 2=Sometimes, 3=Never	1.46
What YP thinks about School	Preferences/Tastes	9	Both	1=Always like it, 2=Sometimes like it, 3=Never like it	1.81
YP Preferred adult occupation 1	Preferences/Tastes	9	Both	Teaching	0.13
YP Preferred adult occupation 2	Preferences/Tastes	9	Both	Health & care	0.11
YP Preferred adult occupation 3	Preferences/Tastes	9	Both	Professional STEM	0.04
YP Preferred adult occupation 4	Preferences/Tastes	9	Both	Technical STEM	0.01
YP Preferred adult occupation 5	Preferences/Tastes	9	Both	Trades/construction/mechanic	0.04
YP Preferred adult occupation 6	Preferences/Tastes	9	Both	Transport/agriculture/production	0.05

YP Preferred adult occupation 7	Preferences/Tastes	9	Both	Arts/mdeia	0.38
YP Preferred adult occupation 8	Preferences/Tastes	9	Both	Sport	0.01
YP Preferred adult occupation 9	Preferences/Tastes	9	Both	Protective services	0.05
YP Preferred adult occupation 10	Preferences/Tastes	9	Both	Business/law/management	0.04
YP Preferred adult occupation 11	Preferences/Tastes	9	Both	Clerical/service admin	0.01
YP Preferred adult occupation 12	Preferences/Tastes	9	Both	Dont know/uncodeable	0.14
YP Practical Favourite Subject	Preferences/Tastes	13	LC	=1 if favourite JC subject is a practical based subject	0.18
YP Own Expectations regarding their education	Preferences/Tastes	13	LC	Educational expectations (1–6)	3.29
Maths Difficult	Preferences/Tastes	13	LC	1=Difficult, 2=OK, 3=Not difficult	2.27
English Difficult	Preferences/Tastes	13	LC	1=Difficult, 2=OK, 3=Not difficult	2.53
Disability status of YP	Background	9	Both	=1 if child has a disability	0.10
Immigrant Status of PCG	Background	9	Both	=1 if parent immigrant	0.15
Number of Books in home	Background	9	Both	Number of books (bands)	4.25
PCG is university level educated	Background	9	Both	=1 if mother has a university degree	0.18

Robustness Testing/Machine Learning - Additional Variables

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Teacher assessment of YP: reading	Comp. Adv/Skills	9	Both	Teacher assessment of child's reading. 1=Below average, 2=average, 3=above average	2.29
Teacher assessment of YP: writing	Comp. Adv/Skills	9	Both	Teacher assessment of child's writing. 1=Below average, 2=average, 3=above average	2.18
Teacher assessment of YP: comprehension	Comp. Adv/Skills	9	Both	Teacher assessment of child's comprehension. 1=Below average, 2=average, 3=above average	2.27
Teacher assessment of YP: maths	Comp. Adv/Skills	9	Both	Teacher assessment of child's maths. 1=Below average, 2=average, 3=above average	2.23
Teacher assessment of YP: creativity	Comp. Adv/Skills	9	Both	Teacher assessment of child's creativity. 1=Below average, 2=average, 3=above average	2.28
Teacher assessment of YP: oral communication	Comp. Adv/Skills	9	Both	Teacher assessment of child's oral communications. 1=Below average, 2=average, 3=above average	2.31

Teacher assessment of YP: problem solving	Comp. Adv/Skills	9	Both	Teacher assessment of child's problem-solving. 1=Below average, 2=average, 3=above average	2.13
PCG perception of YP: above average math	Comp. Adv/Skills	9	Both	=1 if PCG rates child's maths ability as above average or excellent (relative to other children of same age)	0.55
PCG perception of YP: above average reading	Comp. Adv/Skills	9	Both	=1 if PCG rates child's reading ability as above average or excellent (relative to other children of same age)	0.61
BAS Score	Comp. Adv/Skills	13	LC	BAS Score measure	115.19
SDQ Score - Total from PCG questions	Comp. Adv/Skills	13	LC	Non-cognitive measure	7.54
Parent/guardian expectations of YP	Preferences/Tastes	9	Both	How far PCG expects their child will go in education 1=Junior Certificate or equivalent, 2=Leaving Certificate or equivalent, 3=An apprenticeship or trade, 4=Diploma/Certificate, 5=Degree, 6=Postgraduate/higher degree	4.74
Under 1 hour TV per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour watching television each day	0.24
Under 1 hour reading per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour reading each day	0.67
Under 1 hour computer per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour on computer each day	0.86
Under 1 hour gaming per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour on video games each day	0.79
Under 1 hour homework per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour on homework each day	0.74
Computer in bedroom	Preferences/Tastes	9	Both	=1 if child has computer in room	0.07
Games console in bedroom	Preferences/Tastes	9	Both	=1 if child has games console in room	0.32
Sports/fitness club	Preferences/Tastes	9	Both	=1 if during an average week, child participates in sports/fitness club	0.78
Cultural activities	Preferences/Tastes	9	Both	=1 if during an average week, child participates in cultural activities	0.50
Youth club member	Preferences/Tastes	9	Both	=1 if during an average week, child participates in youth club	0.06
Scouts member	Preferences/Tastes	9	Both	=1 if during an average week, child participates in scouts/guides etc.	0.15

	Homework club member	Preferences/Tastes	9	Both	=1 if during an average week, child participates in homework club	0.06
	Own bedroom	Preferences/Tastes	9	Both	=1 if child has their own bedroom	0.67
	Thoughts on school	Preferences/Tastes	9	Both	1=Child always likes school, 2=Sometimes, 3=Never	1.81
	Self: doing well in schoolwork	Preferences/Tastes	9	Both	How does child think they're doing in school work? 1=Well, 2=Average/ok, 3=Poorly	1.37
	Like Irish	Preferences/Tastes	9	Both	1=Child always likes Irish, 2=Sometimes, 3=Never	2.04
	Look forward to school	Preferences/Tastes	9	Both	1=Child always looks forward to school, 2=Sometimes, 3=Never	1.86
	Like teacher	Preferences/Tastes	9	Both	1=Child always likes teacher, 2=Sometimes, 3=Never	1.52
	Top 3 activities: friends	Preferences/Tastes	9	Both	=1 if child rates hanging out with friends as one of top 3 best things to do in spare time	0.65
37	Top 3 activities: chat online	Preferences/Tastes	9	Both	=1 if child rates chatting to friends on phone/computer as one of top 3 best things to do in spare time	0.09
	Top 3 activities: sport	Preferences/Tastes	9	Both	=1 if child rates playing sport as one of top 3 best things to do in spare time	0.62
	Top 3 activities: TV	Preferences/Tastes	9	Both	=1 if child rates watching television as one of top 3 best things to do in spare time	0.30
	Top 3 activities: gaming	Preferences/Tastes	9	Both	=1 if child rates playing video games as one of top 3 best things to do in spare time	0.18
	Top 3 activities: reading	Preferences/Tastes	9	Both	=1 if child rates reading as one of top 3 best things to do in spare time	0.25
	Top 3 activities: outdoors	Preferences/Tastes	9	Both	=1 if child rates playing games outside as one of top 3 best things to do in spare time	0.44
	Top 3 activities: listening to music	Preferences/Tastes	9	Both	=1 if child rates listening to music as one of top 3 best things to do in spare time	0.19
	Top 3 activities: family	Preferences/Tastes	9	Both	=1 if child rates talking to family as one of top 3 best things to do in spare time	0.22

Read daily	Preferences/Tastes	9	Both	=1 if child reads for fun daily	0.32
Least Favourite Subject	Preferences/Tastes	13	LC	=1 if YP report maths or science as their least favourite subject in school	0.20
Maths Interesting	Preferences/Tastes	13	LC	1=Child thinks Math is Interesting, 2=Ok, 3=Not Interesting	1.85
English Interesting	Preferences/Tastes	13	LC	1=Child thinks English is Interesting, 2=Ok, 3=Not Interesting	1.67
Single-sex school	School/Peer Effects	9	Both	=1 if primary school is single-sex	0.46
Transition year Offered in School	School/Peer Effects	17/18	LC	=1 if school offers Transition Year	0.98
Parent/Guardian Disability status	Background	9	Both	=1 if primary caregiver has a disability	0.12
Immigrant Status of YP	Background	9	Both	=1 if child is immigrant	0.11
Dyslexic Status of YP	Background	9	Both	=1 if child has dyslexia	0.03
ADHD Status of YP	Background	9	Both	=1 if child has ADHD	0.00
Autism Status of YP	Background	9	Both	=1 if child has autism	0.01
YP has Speech/language difficulty	Background	9	Both	=1 if child has speech/language difficulties	0.01
Dyspraxia Status of YP	Background	9	Both	=1 if child has dyspraxia	0.01
Slow progress Status of YP	Background	9	Both	=1 if child has slow progress	0.02
Household type 1	Background	9	Both	=1 if Single parent plus 1–2 children	0.09
Household type 2	Background	9	Both	=1 if Single parent plus 3+ children	0.05
Household type 3	Background	9	Both	=1 if Couple plus 1–2 children	0.36
Household type 4	Background	9	Both	=1 if Couple plus 3+ children	0.49
5 or more in household	Background	9	Both	=1 if >4 people in household	0.58
YP is Regularly unwell	Background	9	Both	=1 if child's health rated worse than 'Very healthy, no problems'	0.26
YP has Sight issue	Background	9	Both	=1 if child has/had had sight issue requiring correction	0.16
YP has Hearing issue	Background	9	Both	=1 if child has/had had hearing issue requiring correction	0.08
YP has Speech issue	Background	9	Both	=1 if PCG worries about how child talks and makes speech sounds	0.99
PCG basic calculations	Background	9	Both	=1 if PCG can tell if they have the correct change when buying things with €5 or €10	0.98
Parents meet teacher regularly	Background	9	Both	=1 PCG or spouse attended formal meeting with teacher in last academic year	0.97

Under 3 absent days	Background	9	Both	=1 if child was absent for <3 days in last academic year	0.73
Parents always help with homework	Background	9	Both	=1 if PCG or spouse always/nearly always helps child with homework	0.49

Note: The column labelled “LC or JC” indicates whether a variable is included in the lower secondary (JC) or upper secondary (LC) analysis. For the lower secondary analysis, explanatory variables are restricted to measures collected in Wave 1 (age 9), meaning that only age-9 variables are common to both analyses. Variables collected in Wave 2 or Wave 3 (ages 13 and 17/18, respectively) are used exclusively in the upper secondary analysis.

VI.4 EXTREME GRADIENT BOOSTING

Students’ decisions are influenced by a wide range of personal and school-level characteristics that interact in non-obvious, often nonlinear ways. Standard regression techniques can be poorly suited to this context: they require prior specification of which variables matter and how they relate to the outcome, assumptions that are difficult to justify with such rich data. Moreover, including many predictors in a linear model introduces the risk of collinearity, overfitting, and loss of clarity in interpretation. To navigate these challenges, we adopt the machine learning method eXtreme Gradient Boosting (XGBoost) (Chen and Guestrin, 2016), which accommodates complex interactions and nonlinearities without predefining relationships between variables.

XGBoost constructs a predictive model by building a sequence of decision trees, each improving on the previous one. The final prediction is a weighted sum of individual tree contributions. It works by identifying patterns in the training data (comprising 70% of our sample) and using them to forecast STEM subject choices.⁹ Each tree divides the data based on values of predictor variables, grouping students with similar likelihoods of studying STEM. While individual decision trees can overfit the training data, XGBoost avoids this by combining many shallow trees, each correcting the mistakes of the previous ones. This process allows it to capture subtle relationships while avoiding the pitfalls of more flexible but less disciplined models.

Hyperparameters are selected to balance flexibility with generalisability. In particular, *max_depth* controls how deep each decision tree may grow. This hyperparameter is tuned by cross-validation; we search over values $\{3, 4, 5, 6, 7\}$ using 5-fold cross-validation on the training data, selecting the value that maximises the area under the ROC curve (AUC). All remaining hyperparameters follow standard applied practice to prevent overfitting: the learning rate is $\eta = 0.05$; the number of boosting rounds is fixed at *nrounds* = 100; each tree uses a random 80% subsample of the training observations (*subsample* = 0.8) and a random 80% subset of predictors (*colsample_bytree* = 0.8); and $\gamma = 0$, *min_child_weight* = 1. Prior to fitting, missing values are handled by median imputation for continuous variables and modal-category imputation for categorical variables.¹⁰

Both models achieve similar levels of predictive accuracy (approximately 77–78%) and outperform the no-information rate, indicating meaningful predictive performance. However, the JC STEM model demonstrates a higher AUC (0.86 versus 0.80) and substantially greater sensitivity, suggesting it is more effective at identifying STEM participants. The LC STEM model achieves higher specificity, indicating stronger performance in correctly classifying non-STEM outcomes. Overall, the JC model attains a higher F1 score and lower Brier score, pointing to a more balanced and better-calibrated model, whereas the LC model

Table A.5: Predictive Performance of XGBoost Models

Model	NIR	Test (Acc)	Test (AUC)	Sensitivity	Specificity	Precision	F1 Score	Brier
JC	0.65	78.23	0.86	64.32	85.79	71.11	67.55	0.144
LC	0.71	77.43	0.80	43.64	91.38	67.61	53.04	0.158

Note: This table reports out-of-sample predictive performance for the XGBoost models used in the SHAP decomposition analysis. NIR denotes the no-information rate. Sensitivity measures the share of true STEM participants correctly classified, while specificity measures the share of non-STEM students correctly classified. The JC model exhibits a higher AUC, F1 score, and lower Brier score, indicating more balanced predictive performance overall, whereas the LC model achieves higher specificity but lower sensitivity.

appears more conservative, prioritising correct classification of the majority class.

VI.5 SHAP DECOMPOSITIONS

To fully capitalise on the flexible, non-linear nature of the XGBoost model, we use SHAP (Shapley Additive exPlanations) values for decomposition analysis. SHAP values quantify each predictor’s contribution to the predicted probability of STEM enrolment, satisfying fairness criteria derived from cooperative game theory.

The decomposition is performed on the held-out test set (the 30% of observations not used in training). For each predictor x_i , we compute its weighted mean SHAP value separately for female and male students. The contribution of variable x_i to the predicted gender gap is:

$$\bar{\phi}_i^M - \bar{\phi}_i^F$$

where $\bar{\phi}_i^M$ and $\bar{\phi}_i^F$ are the weighted mean SHAP values for male and female students respectively. Summing these contributions across all predictors recovers the total predicted gender gap. Individual predictors are then aggregated into the variable groups defined in Table A.4 (STEM-readiness, comparative advantage, peers and school factors, preferences and tastes, and background characteristics). The SHAP value attributed to the gender indicator itself captures the residual direct contribution of gender that is not explained by the other included variables.

Traditional decomposition methods typically dissect the observed enrolment gap. In contrast, our SHAP-based decomposition analyses the gap predicted by the XGBoost model, fully utilising all available predictors and their interactions without restrictive assumptions such as linearity or preselected variables. While traditional decomposition methods rely heavily on linearity assumptions, SHAP decomposition does not. However, the validity of the SHAP decomposition hinges on the accuracy of the underlying model predictions. If the model systematically mispredicts enrolment probabilities differently by gender, the decomposed

predicted gap may differ from the true observed gap. Therefore, our results section clearly distinguishes between the observed and predicted gaps, enabling a transparent assessment of potential discrepancies.

VI.6 REGRESSION RESULTS - LC & JC

Table A.6: Determinants of LC Practical STEM Participation

	Coef	SE
Male	0.071**	(0.034)
<i>Group 1 – STEM Readiness/Junior Cert</i>		
Studied Foreign Language for JC	-0.028	(0.040)
Studied Arts for JC	-0.044*	(0.024)
Studied Music for JC	-0.039	(0.024)
Studied Home Economics for JC	-0.030	(0.025)
Studied Material Technology for JC	0.110***	(0.034)
Studied Metalwork for JC	0.141***	(0.041)
Studied Tech Graph for JC	0.230***	(0.031)
Studied Technology for JC	0.198***	(0.052)
Studied Business Studies for JC	-0.036	(0.023)
Studied Religion for JC	-0.005	(0.021)
Studied Higher-Level English for JC	-0.005	(0.045)
Studied Higher-Level Maths for JC	-0.059*	(0.032)
Did Transition Year in School	0.040*	(0.021)
<i>Group 2 – Comparative Advantage/Skills</i>		
Drumcondra Numerical Score	-0.012	(0.012)
Drumcondra Reading Score	-0.031**	(0.013)
Academic Self Concept Score	-0.000	(0.003)
Conscientious Score	-0.007	(0.004)
SDQ Score	-0.002	(0.002)
Number of Higher Level sat for JC	0.013**	(0.006)
JC English Grade	-0.004	(0.010)
JC Maths Grade	-0.004	(0.009)
<i>Group 3 – Peer Effects/School Factors</i>		
<i>School Gender (ref: Girls single-sex)</i>		
Boys – Single Sex	0.026	(0.057)
Mixed Sex	0.016	(0.042)
<i>School Type (ref: Vocational)</i>		
Fee-paying	-0.004	(0.050)
Non-fee paying	-0.025	(0.040)
Community	-0.031	(0.036)
DEIS status of school	0.003	(0.026)
French Offered in School	-0.070	(0.095)
German Offered in School	0.016	(0.023)
Spanish Offered in School	-0.005	(0.021)
Arts Offered in School	-0.009	(0.042)
Applied Maths Offered in School	-0.015	(0.021)
Physics Offered in School	0.041	(0.045)

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Table A.6 continued

	Coef	SE
Chemistry Offered in School	-0.009	(0.073)
Agricultural Science Offered in School	0.086***	(0.021)
Biology Offered in School	0.211***	(0.074)
Engineering Offered in School	0.023	(0.036)
Construction Offered in School	-0.003	(0.036)
Accounting Offered in School	0.012	(0.037)
Business Offered in School	-0.046	(0.058)
Economics Offered in School	-0.012	(0.024)
Technology Offered in School	0.013	(0.026)
Music Offered in School	-0.003	(0.033)
Home Economics Offered in School	0.063	(0.039)
Project Maths Offered in School	0.012	(0.023)
Religious Studies Offered in School	-0.009	(0.019)
Link Modules Offered in School	-0.003	(0.020)
DCG Offered in School	0.014	(0.030)
Other Languages Offered in School	-0.051	(0.041)
<i>Group 4 – Preferences/Tastes</i>		
<i>Like Maths (ref: Always)</i>		
Like Maths (Sometimes)	0.022	(0.019)
Like Maths (Never)	-0.080**	(0.031)
<i>Like Reading (ref: Always)</i>		
Like Reading (Sometimes)	0.016	(0.018)
Like Reading (Never)	-0.026	(0.047)
Practical Favourite Subject	0.146***	(0.034)
<i>Own Expectations of educational attainment (ref: JC)</i>		
Own Expectations (LC)	-0.052	(0.060)
Own Expectations (Higher Certificate)	-0.025	(0.056)
Own Expectations (Degree or higher degree)	-0.062	(0.057)
<i>Maths Difficult (ref: Difficult)</i>		
Maths Difficult (Ok)	0.032	(0.028)
Maths Difficult (Not Difficult)	0.042	(0.032)
<i>English Difficult (ref: Difficult)</i>		
English Difficult (Ok)	0.021	(0.049)
English Difficult (Not Difficult)	-0.002	(0.048)
<i>Like School (ref: Always)</i>		
Like School (Sometimes)	-0.014	(0.022)
Like School (Never)	0.035	(0.042)
<i>Occupation (ref: Teaching)</i>		
YP Preferred adult occupation – Healthcare	-0.005	(0.033)
YP Preferred adult occupation – Professional STEM	0.084	(0.064)
YP Preferred adult occupation – Technical STEM	-0.112	(0.177)
YP Preferred adult occupation – Trades/construction/mechanics	0.085	(0.052)
YP Preferred adult occupation – Transport/agricultural/production	0.102**	(0.052)
YP Preferred adult occupation – Arts/Media	0.012	(0.028)
YP Preferred adult occupation – Sports	-0.147***	(0.040)
YP Preferred adult occupation – Protective Services	-0.003	(0.046)
YP Preferred adult occupation – Business/law/management	-0.038	(0.052)
YP Preferred adult occupation – Clerical/service admin	0.090	(0.129)

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Table A.6 continued

	Coef	SE
YP Preferred adult occupation – Other/unknown	0.002	(0.030)
<i>Group 5 – Background</i>		
YP has a Disability	0.005	(0.030)
PCG is born outside of Ireland	-0.072***	(0.026)
PCG is University level Educated	0.010	(0.023)
<i>Books in Home (ref: None)</i>		
Number of Books in Home (<10)	0.332	(0.290)
Number of Books in Home (10-20)	0.333	(0.288)
Number of Books in Home (21-30)	0.336	(0.289)
Number of Books in Home (>30)	0.342	(0.289)
R-Squared	0.30	
Observations	3,888	

Note: Linear probability model estimates of participation in practical STEM subjects at Leaving Certificate level. The dependent variable is an indicator equal to one if the student takes at least one practical STEM subject and zero otherwise. Robust standard errors clustered by school in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.7: Determinants of JC Practical STEM Participation

	Coef	SE
Male	0.570***	(0.033)
<i>Group 1 – Comparative Advantage</i>		
Drumcondra Numerical Score	-0.010	(0.012)
Drumcondra Reading Score	-0.046***	(0.014)
<i>Group 2 – Preferences</i>		
<i>Like Maths (ref: Always)</i>		
Like Maths (Sometimes)	-0.000	(0.019)
Like Maths (Never)	-0.047	(0.030)
<i>Like Reading (ref: Always)</i>		
Like Reading (Sometimes)	-0.014	(0.019)
Like Reading (Never)	-0.066	(0.046)
<i>Like School (ref: Always)</i>		
Like School (Sometimes)	0.019	(0.020)
Like School (Never)	0.068*	(0.038)
<i>Occupation (ref: Teaching)</i>		
YP Preferred adult occupation – Healthcare	-0.015	(0.031)
YP Preferred adult occupation – Professional STEM	0.072	(0.058)
YP Preferred adult occupation – Technical STEM	0.233*	(0.138)
YP Preferred adult occupation – Trades/construction/mechanics	0.101**	(0.047)
YP Preferred adult occupation – Transport/agricultural/production	0.050	(0.038)
YP Preferred adult occupation – Arts/Media	0.028	(0.026)
YP Preferred adult occupation – Sports	-0.083	(0.081)
YP Preferred adult occupation – Protective Services	0.006	(0.045)
YP Preferred adult occupation – Business/law/management	0.030	(0.061)
YP Preferred adult occupation – Clerical/service admin	-0.086	(0.079)
YP Preferred adult occupation – Other/unknown	0.045	(0.033)

Continued on next page

Table A.7 continued

	Coef	SE
<i>Group 3 – Background</i>		
YP has a Disability	-0.036	(0.030)
PCG is born outside of Ireland	0.038	(0.024)
PCG is University level Educated	-0.079***	(0.029)
<i>Books in Home (ref: None)</i>		
Number of Books in Home (<10)	-0.050	(0.127)
Number of Books in Home (10-20)	-0.050	(0.125)
Number of Books in Home (21-30)	-0.042	(0.125)
Number of Books in Home (>30)	-0.056	(0.123)
R-Squared	0.37	
Observations	3,888	

Note: Linear probability model estimates of participation in practical STEM subjects at Junior Certificate level. The dependent variable is an indicator equal to one if the student takes at least one practical STEM subject and zero otherwise. Robust standard errors clustered by school in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

VI.7 UPPER AND LOWER SECONDARY GENDER GAP - BY INDIVIDUAL SUBJECT

Appendix Table A.8 extends the analysis by examining the gender gap separately across individual practical STEM subjects at the Leaving Certificate level: Technology, Construction Studies, Engineering, Design and Communication Graphics (DCG), and Agricultural Science. In the baseline specification, substantial gender differences are observed across all subjects. The largest disparities appear in Construction Studies and Engineering, where the female coefficients are 20.40 and 13.47 percentage points, respectively, while smaller but still sizeable gaps exist in Technology (4.52), DCG (12.00), and Agricultural Science (4.85). Introducing the full set of covariates substantially reduces the estimated gender differences in several subjects. In Technology and DCG, the gender gap is almost entirely eliminated and becomes statistically insignificant. In Construction Studies and Engineering, the gender disparity remains statistically significant but is considerably smaller in magnitude. Agricultural Science also shows a large reduction in the gender coefficient once observable characteristics are included.

As in the main analysis, Panel B reports the decomposition of these changes. Across subjects, the dominant contributor to the reduction in the gender coefficient is STEM-readiness, captured primarily by Junior Cycle subject choices and related indicators. For Construction Studies, STEM-readiness accounts for nearly all of the reduction in the gender gap, explaining 11.94 percentage points of the total change. Similar patterns emerge in Technology and DCG, where earlier subject choices explain the majority of the observed reduction in the gender coefficient. In Engineering, STEM-readiness accounts for the largest share of the change, though additional mechanisms also play a role. Even in Agricultural Science, where the overall

gender gap is smaller, earlier subject preparation explains most of the reduction observed when controls are introduced.

Other mechanisms play a more limited and heterogeneous role across subjects. Measures of comparative advantage contribute very little to explaining the gender gap across the subjects considered. Peer effects enter with a negative sign in Construction Studies and Engineering, indicating they widen rather than narrow the explained portion of the gap. Preference-related variables also explain a non-trivial share of the change in several subjects, particularly Construction Studies, Engineering, and DCG, although their contribution is smaller than that of STEM-readiness. Background characteristics have minimal explanatory power across all subjects.

Taken together, these subject-level results reinforce the pipeline interpretation observed in the main analysis. Much of the gender disparity in practical STEM participation at the end of secondary school can be traced to earlier differences in curricular pathways established during lower secondary education. Subject choices made at the Junior Cycle stage appear to play a central role in shaping access to practical STEM subjects later in the educational trajectory.

Table A.8: LC Practical STEM — Gender gap and Gelbach decomposition - By Individual Subject

	Technology	Construction	Engineering	DCG	Ag Science
<i>Panel A: OLS estimates</i>					
Baseline - Female (1)	4.52*** (0.01)	20.40*** (0.02)	13.47*** (0.01)	12.00*** (0.01)	4.85*** (0.02)
Full Model - Female (2)	0.22 (0.01)	8.43*** (0.02)	8.09*** (0.02)	-0.97 (0.02)	0.78 (0.02)
Observations	3,888	3,888	3,888	3,888	3,888
<i>Panel B: Gelbach decomposition of the change in the gender gap</i>					
Total change: (1)–(2)	4.30*** (0.02)	11.97*** (0.02)	5.38*** (0.02)	12.97*** (0.02)	4.07 (0.02)
<i>Contribution to change:</i>					
STEM-readiness	2.92*** (0.02)	11.94*** (0.02)	6.42*** (0.02)	8.49*** (0.02)	3.27** (0.02)
Comparative advantage	0.21 (0.00)	0.20 (0.00)	0.08 (0.00)	0.00 (0.00)	0.09 (0.00)
Peer effects	0.20 (0.01)	-3.00** (0.01)	-3.64*** (0.01)	1.20 (0.01)	0.00 (0.01)
Preferences	1.10** (0.01)	3.22*** (0.01)	2.39*** (0.01)	3.30*** (0.01)	0.65 (0.01)
Background	-0.19* (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	-0.01 (0.00)

Note: Robust standard errors clustered by school in parentheses. Statistical significance is at: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Panel B reports a Gelbach (2016) decomposition of the difference in the Female coefficient between the baseline and full specification, allocating the total change across pre-defined covariate groups.

Appendix Table A.9 reports the subject-level results for practical STEM participation at the Junior Cycle stage. Large gender disparities are evident across all subjects in the baseline specification. The largest gap appears in Material Technology, where the female coefficient is 36.02 percentage points, followed by Technical Graphics (27.33) and Metalwork (16.29). Technology exhibits a smaller but still notable gender gap of -9.00 percentage points.

Introducing the full set of age-9 covariates leads to only modest changes in the estimated gender coefficients. For Technology, the coefficient falls slightly from 9.00 to 8.49 percentage points. Material Technology and Metalwork show marginally larger reductions, from 36.02 to 32.96 and from 16.29 to 14.56, respectively, though substantial gender gaps remain. In Technical Graphics, the estimated female coefficient becomes slightly larger in magnitude, moving from 27.33 to 28.27 percentage points. Overall, the inclusion of observable characteristics measured prior to secondary school entry explains only a small fraction of the gender differences in subject participation.

Consistent with the aggregate Junior Cycle results, the explanatory power of the observed variables is

limited across subjects. Preference-related variables account for the largest share of the explained component in Material Technology and Metalwork, contributing 3.38 and 1.99 percentage points, respectively, to the reduction in the gender coefficient. However, these effects remain small relative to the overall gender gaps observed in these subjects.

Measures of comparative advantage contribute very little to explaining the observed disparities. Their estimated contributions are small in magnitude and, across subjects, generally statistically insignificant. Background characteristics likewise account for almost none of the change in the gender coefficients.

Once again, the subject-level results reinforce the broader conclusion that the gender gap in practical STEM participation at the Junior Cycle stage cannot be explained by differences observable at age 9. The mechanisms captured in our data account for only a very small share of the substantial gender disparities observed across practical STEM subjects.

Table A.9: JC Practical STEM — Gender gap and Gelbach decomposition - By Individual Subject

	Technology	Material Technology	Metalwork	Tech Graph
<i>Panel A: OLS estimates</i>				
Baseline - Female (1)	9.00*** (0.02)	36.02*** (0.03)	16.29*** (0.02)	27.33*** (0.02)
Full Model - Female (2)	8.49*** (0.02)	32.96*** (0.03)	14.56*** (0.02)	28.27*** (0.03)
Observations	3,888	3,888	3,888	3,888
<i>Panel B: Gelbach decomposition of the change in the gender gap</i>				
Total change: (1)–(2)	0.51 (0.01)	3.07*** (0.01)	1.73*** (0.01)	-0.94 (0.01)
<i>Contribution to change:</i>				
Comparative advantage	0.07 (0.00)	-0.52 (0.01)	-0.51** (0.00)	0.29 (0.00)
Preferences	0.69 (0.01)	3.38*** (0.01)	1.99*** (0.01)	-0.97 (0.01)
Background	-0.25 (0.00)	0.21 (0.00)	0.25 (0.00)	-0.26 (0.00)

Note: Robust standard errors clustered by school in parentheses. Statistical significance is at: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Panel B reports a Gelbach (2016) decomposition of the difference in the Female coefficient between the baseline and full specification, allocating the total change across pre-defined covariate groups.